THE SEPARATION AXIOMS FOR MORPHISMS OF FORMAL ALGEBRAIC SPACES ARE STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one chain of morphisms; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The separation axioms are defined by requiring the diagonal to have a property. Same chapter, Definition definition-separated-morphism, invoked in the statement and at proof lines 6874-6876. Reduces each axiom to a property of the diagonal.
- The base changed map in the factorisation of the diagonal has the required property. Same chapter, Lemma lemma-fibre-product-after-map, invoked at proof lines 6876-6878. Gives the property for the second factor of the diagonal of the composite.

Source reference. Original lemma 0ARQ: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 763–763. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over S. Let $\Delta_{X/Y} : X \to X \times_Y X$ be the diagonal morphism.

(1) We say f is *separated* if $\Delta_{X/Y}$ is a closed immersion.
(2) We say f is *quasi-separated* if $\Delta_{X/Y}$ is quasi-compact.

Lemma 0.2. *All of the separation axioms listed in Definition 0.1 are stable under composition of morphisms.*

Proof. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of formal algebraic spaces to which the axiom in question applies. The diagonal $\Delta_{X/Z}$ is the composition

$$X \longrightarrow X \times_Y X \longrightarrow X \times_Z X.$$

Our separation axiom is defined by requiring the diagonal to have some property $\mathcal{P}$. By Lemma 0ARQ above we see that the second arrow also has this property.

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Hence the lemma follows since the composition of (representable) morphisms with property $\mathcal{P}$ also is a morphism with property $\mathcal{P}$. $\square$